Firms in Competitive Markets

EC 111 Principles of Microeconomics

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Reading

- These slides are not a substitute for reading Mankiw, Chapter 15, "Firms in Competitive Markets".
- Required reading: Mankiw, *Principles of Microeconomics*, 11th edition, Chapter 15, "Firms in Competitive Markets".

What you must know from this chapter

1. Overview of market structures.
2. Conditions necessary to achieve perfect competition.
3. Profit maximization (MR = MC).
4. Graphical representation of profits/losses.
5. Shut-down point – short-run and long-run.
6. Short-run supply curve.
7. Entry/exit.
8. Firm and industry equilibrium and changes to equilibrium.

Part 1

Market Structures and Price Taking

Four market structures

Definition: Market structure

The number of firms, whether their products are identical or differentiated, and how easy it is to enter or leave the market.

Structure	Main feature	Firm's pricing position
Perfect competition	Many sellers of identical goods	Price taker
Monopolistic competition	Many sellers of differentiated goods	Some say over price
Oligopoly	A few important competing firms	Decisions depend on rivals
Monopoly	One seller; no close substitutes; barriers	Faces market demand

Perfect competition

Definition: Perfectly competitive market

- Many buyers and many sellers.
- Firms sell **identical** products.
- Each buyer and each seller is too small to move the market price.
- In the long run, firms can enter and leave freely: no significant barriers, legal or otherwise.

Why a competitive firm is a price taker

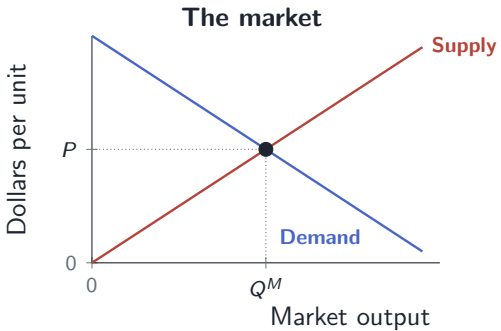
If products are identical and there are many sellers, **then**:

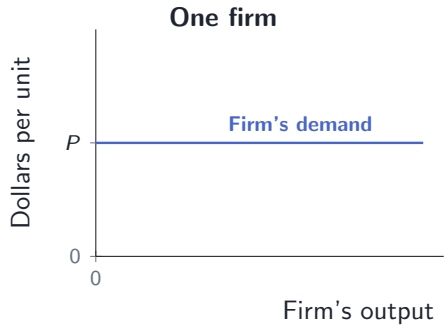
- A firm that charges *above* the market price sells nothing. Buyers switch to identical goods from other sellers.
- A firm that charges *below* the market price gains nothing. It can already sell all it wants at the market price.
- One firm's output is too small to change the market price.

Finding

The firm takes the market price as given. Its demand curve is horizontal at that price.

Market demand versus the firm's demand





Finding

Market demand slopes down; the market sets P. One firm's demand is flat at P. If the market price changes, the firm's flat line moves with it.

Part 2

Price, Average Revenue and Marginal Revenue

Why $P = \text{AR} = \text{MR}$ for a price taker

Definition: Revenue

Total revenue $\text{TR} = P \times Q$. Average revenue $\text{AR} = \text{TR} \div Q$. Marginal revenue $\text{MR} = \Delta\text{TR}/\Delta Q$: the extra revenue from one more unit.

- $\text{AR} = \dfrac{P \times Q}{Q} = P$. Every unit sells for the same price.
- One more unit adds P to revenue, because the price does not fall when the firm sells more: $\Delta\text{TR} = P \times \Delta Q$, so $\text{MR} = P$.

Finding

For a price taker, $P = \text{AR} = \text{MR}$. The firm's demand line is also its AR and MR line.

Example: a price of \$10

Q	$\text{TR} = 10 \times Q$	$\text{AR} = \text{TR} \div Q$	$\text{MR} = \Delta\text{TR}/\Delta Q$
1	$10 \times 1 = \$10$	$10 \div 1 = \$10$	$(10 - 0)/1 = \$10$
2	$10 \times 2 = \$20$	$20 \div 2 = \$10$	$(20 - 10)/1 = \$10$
3	$10 \times 3 = \$30$	$30 \div 3 = \$10$	$(30 - 20)/1 = \$10$
4	$10 \times 4 = \$40$	$40 \div 4 = \$10$	$(40 - 30)/1 = \$10$

Output rises one unit at a time, so $\Delta Q = 1$. Price, AR and MR are all \$10 in every row.

Part 3

Profit Maximization: MR = MC

Economic profit

Definition: Economic profit

$$\pi = \text{TR} - \text{TC}$$

- Total revenue and total cost both depend on output, Q:

$$\pi = \text{TR}(Q) - \text{TC}(Q)$$

- So profit depends on output too:

$$\pi(Q) = \text{TR}(Q) - \text{TC}(Q)$$

When is it worth producing more?

Raise output by ΔQ. Profit changes by

$$\Delta\pi = \Delta\text{TR} - \Delta\text{TC}.$$

- If $\Delta\text{TR} > \Delta\text{TC}$, then $\Delta\pi > 0$: profit rises. It is worth increasing output.
- If $\Delta\text{TR} < \Delta\text{TC}$, then $\Delta\pi < 0$: profit falls. It is not worth increasing output.

Finding

Keep increasing output as long as the extra revenue exceeds the extra cost. Stop when it no longer does.

The rule, derived

Definition: Marginal cost (from the Costs lecture)
$\text{MC} = \Delta\text{TC}/\Delta Q$: the extra cost of one more unit.

- One more unit changes profit by $\Delta\pi = \Delta\text{TR} - \Delta\text{TC} = \text{MR} - \text{MC}$.
- If $\text{MR} > \text{MC}$, the unit adds to profit: make it. If $\text{MR} < \text{MC}$, it subtracts: do not.
- MC rises as output grows (diminishing marginal product). So the firm keeps adding units until MC reaches MR.

Finding
Produce where $\text{MR} = \text{MC}$, on the rising part of MC. For a price taker $\text{MR} = P$, so produce where $P = \text{MC}$.

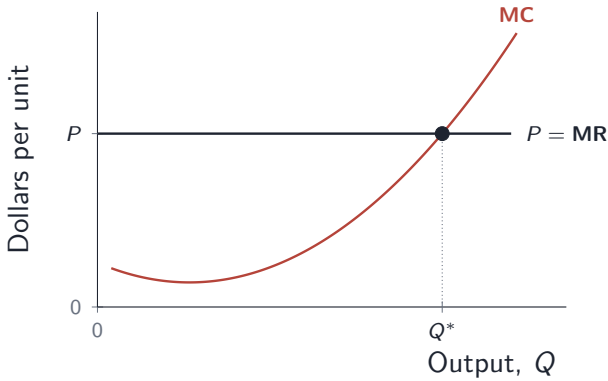
The rule on a graph

The firm faces the flat line $P = \text{MR}$.

Left of Q^*: $P > \text{MC}$. Each extra unit adds profit.

Right of Q^*: $P < \text{MC}$. Each extra unit loses profit.

So profit is greatest at Q^*, where $P = \text{MC}$.



Example: a firm's costs

Total cost is fixed cost plus variable cost: $\text{TC}(Q) = F + \text{VC}(Q)$. Here $F = \$56$ and $\text{VC}(Q) = Q^3 - 12Q^2 + 60Q$.

Q	$\text{VC} = Q^3 - 12Q^2 + 60Q$	$\text{TC} = 56 + \text{VC}$	$\text{MC} = \Delta\text{TC}/\Delta Q$
0	$0 - 0 + 0 = \$0$	$56 + 0 = \$56$	—
1	$1 - 12 + 60 = \$49$	$56 + 49 = \$105$	$(105 - 56)/1 = \$49$
2	$8 - 48 + 120 = \$80$	$56 + 80 = \$136$	$(136 - 105)/1 = \$31$
3	$27 - 108 + 180 = \$99$	$56 + 99 = \$155$	$(155 - 136)/1 = \$19$
4	$64 - 192 + 240 = \$112$	$56 + 112 = \$168$	$(168 - 155)/1 = \$13$
5	$125 - 300 + 300 = \$125$	$56 + 125 = \$181$	$(181 - 168)/1 = \$13$
6	$216 - 432 + 360 = \$144$	$56 + 144 = \$200$	$(200 - 181)/1 = \$19$
7	$343 - 588 + 420 = \$175$	$56 + 175 = \$231$	$(231 - 200)/1 = \$31$
8	$512 - 768 + 480 = \$224$	$56 + 224 = \$280$	$(280 - 231)/1 = \$49$
9	$729 - 972 + 540 = \$297$	$56 + 297 = \$353$	$(353 - 280)/1 = \$73$

$\Delta Q = 1$ in every row. MC first falls, then rises: the U shape from the Costs lecture.

Example: the best output at a price of \$40

Q	TC	MC	TR $= 40 \times Q$	Profit $=$ TR $-$ TC
5	\$181	\$13	\$200	$200 - 181 = \$19$
6	\$200	\$19	\$240	$240 - 200 = \$40$
7	\$231	\$31	\$280	$280 - 231 = \$49$
8	\$280	\$49	\$320	$320 - 280 = \$40$
9	\$353	\$73	\$360	$360 - 353 = \$7$

MR $= P =$ \$40.

The 7th unit costs \$31, below \$40: make it.

The 8th costs \$49, above \$40: stop.

Best output 7. Profit \$49, the highest in the column.

Why fixed cost drops out

Total cost has two parts: $\mathrm{TC}(Q) = F + \mathrm{VC}(Q)$. F has no Q in it.

- Profit: $\pi(Q) = \mathrm{TR}(Q) - F - \mathrm{VC}(Q)$.
- One more unit: $\Delta\pi = \Delta\mathrm{TR} - \Delta\mathrm{VC} - \Delta F$.
- F does not change with output, so $\Delta F = 0$:

$$\Delta\pi = \Delta\mathrm{TR} - \Delta\mathrm{VC} = \mathrm{MR} - \mathrm{MC}.$$

Finding

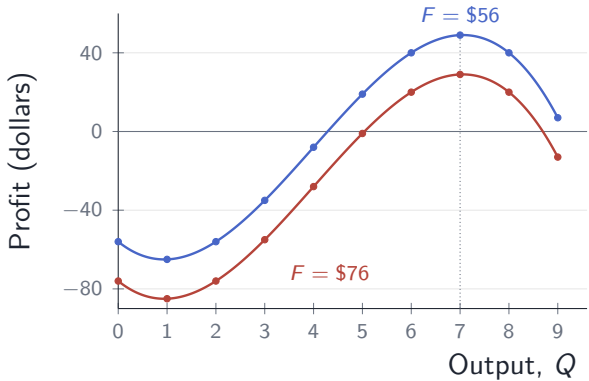
Fixed cost is not in the decision rule, so it cannot change the best output. It changes only how much profit the firm makes.

A higher fixed cost lowers profit, not output

Profit at each output when $P = \$40$ and $F = \$56$.

Fixed cost rises by \$20, to \$76. Every point falls by \$20.

Both curves peak at 7 units. Profit falls from \$49 to \$29.



Your turn: a price of $50

The price rises to $50. Costs are unchanged.

1. What is MR?
2. Should the firm make the 8th unit? The 9th?
3. What is the best output, and the profit there?

Q	VC	TC	MC
0	$0	$56	—
1	$49	$105	$49
2	$80	$136	$31
3	$99	$155	$19
4	$112	$168	$13
5	$125	$181	$13
6	$144	$200	$19
7	$175	$231	$31
8	$224	$280	$49
9	$297	$353	$73

Solution: a price of \$50

1. MR $= P =$ \$50.
2. 8th: MC \$49, below \$50: make it. 9th: MC \$73, above \$50: do not.
3. 8 units. Profit $= 50 \times 8 - 280 = 400 - 280 =$ \$120.

A higher price moves the firm up its MC curve to a larger output.

Q	VC	TC	MC
0	\$0	\$56	—
1	\$49	\$105	\$49
2	\$80	\$136	\$31
3	\$99	\$155	\$19
4	\$112	\$168	\$13
5	\$125	\$181	\$13
6	\$144	\$200	\$19
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Part 4

Profit and Loss on the Graph

Profit as a rectangle

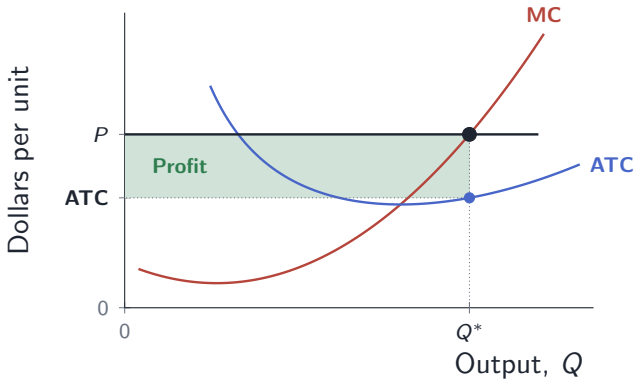
Definition: Average total cost
ATC $=$ TC $\div$ Q: cost per unit.

- TR $= P \times Q$ and TC $=$ ATC $\times$ Q.
- So $\pi =$ TR $-$ TC $= (P -$ ATC$) \times Q$.
- In words: profit per unit, $P -$ ATC, times the number of units. On a graph it is a rectangle: height $P -$ ATC, width Q.

Three cases on one graph

Find Q^* where $P = \text{MC}$. Read ATC at Q^*.

Profit: P is above ATC at Q^*. Rectangle $(P - \text{ATC}) \times Q^*$.



Finding

MR = MC decides how much to make. The gap between P and ATC decides whether that output makes a profit.

Example: the same firm at three prices

Best output from the MC column, then $\pi = \text{TR} - \text{TC}$:

- $P = \$40$: 7 units. $\pi = 280 - 231 = \$49$. ATC $= 231 \div 7 = \$33$, below P. Profit.
- $P = \$33$: 7 units (7th MC \$31; 8th MC \$49). $\pi = 231 - 231 = \$0$. ATC $= \$33 = P$. Zero profit.
- $P = \$30$: 6 units (6th MC \$19; 7th MC \$31). $\pi = 180 - 200 = -\$20$. Loss.

\$33 is the lowest ATC in the table: the break-even price.

Q	VC	TC	MC
0	\$0	\$56	—
1	\$49	\$105	\$49
2	\$80	\$136	\$31
3	\$99	\$155	\$19
4	\$112	\$168	\$13
5	\$125	\$181	\$13
6	\$144	\$200	\$19
7	\$175	\$231	\$31
8	\$224	\$280	\$49
9	\$297	\$353	\$73

Part 5

Shutdown in the Short Run, Exit in the Long Run

Shutdown, exit and sunk cost

Definition: Shutdown

A short-run decision to produce nothing for now. Fixed costs must still be paid.

Definition: Exit

A long-run decision to leave the market. All costs, including fixed costs, stop.

Definition: Sunk cost

A cost already committed that cannot be recovered. It should not affect what the firm does next.

The shutdown rule, derived

In the short run the firm owes its fixed cost (FC) whatever it does.

- Produce: $\pi = \text{TR} - \text{VC} - \text{FC}$.
- Shut down: $\pi = -\text{FC}$.
- FC is in both, so it cannot decide the choice. Producing is worse only if $\text{TR} < \text{VC}$.
- Divide both sides by Q: $\text{TR} \div Q = P$ and $\text{VC} \div Q = \text{AVC}$.

Finding

Short run: shut down if $P < \text{AVC}$. The shutdown price is minimum AVC.

Example: open tomorrow, or shut?

Rent of \$60 is owed tomorrow either way. Operating costs (materials, staff) are \$110.

- Revenue \$90: open $90 - 110 - 60 = -\$80$; shut $-\$60$. Shut: \$90 does not cover the \$110.
- Revenue \$130: open $130 - 110 - 60 = -\$40$; shut $-\$60$. Open: \$130 covers the \$110 and puts \$20 toward rent.

Finding

A firm can rationally operate at a loss. What matters in the short run is whether revenue covers variable cost.

The exit rule, derived

- In the long run, leaving the market avoids every cost, including FC.
- So staying is worse if TR $<$ TC.
- Divide by Q: $P <$ ATC.

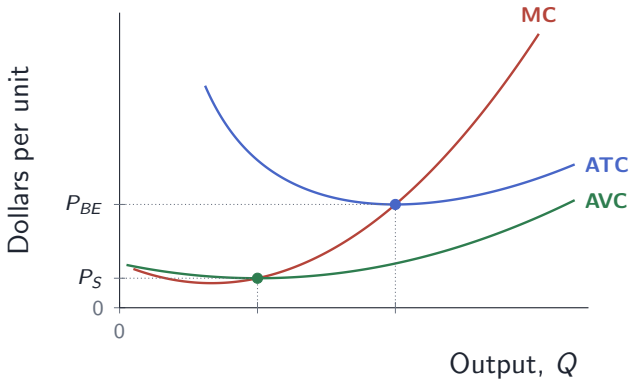
Finding

Long run: exit if $P <$ ATC. The break-even price is minimum ATC.

Shutdown and break-even prices on the graph

MC crosses AVC at its lowest point: the shutdown price P_S.

MC crosses ATC at its lowest point: the break-even price P_{BE}.



Finding

P below minimum AVC: shut down now. P between minimum AVC and minimum ATC: produce at a loss for now; exit if it persists. P above minimum ATC: profit.

Your turn: produce or shut down?

Fixed cost is \$56, paid either way.

1. $P = \$30$. Best output is 6. Produce or shut down?
2. $P = \$24$. Compare producing 6 units with shutting down.
3. $P = \$20$. Produce or shut down?

Q	VC	TC	MC
0	\$0	\$56	—
1	\$49	\$105	\$49
2	\$80	\$136	\$31
3	\$99	\$155	\$19
4	\$112	\$168	\$13
5	\$125	\$181	\$13
6	\$144	\$200	\$19
7	\$175	\$231	\$31
8	\$224	\$280	\$49
9	\$297	\$353	\$73

Solution: produce or shut down?

1. TR $= 30 \times 6 = \$180$, above VC \$144. Produce: loss $180 - 200 = -\$20$, smaller than the \$56 lost by shutting.
2. TR $= 24 \times 6 = \$144$, equal to VC \$144. Loss \$56 either way. \$24 is minimum AVC $(144 \div 6)$: the shutdown price.
3. At every output TR is below VC: AVC is never below \$24, so a \$20 price cannot cover it. Shut down.

Q	VC	TC	MC
0	\$0	\$56	—
1	\$49	\$105	\$49
2	\$80	\$136	\$31
3	\$99	\$155	\$19
4	\$112	\$168	\$13
5	\$125	\$181	\$13
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Part 6

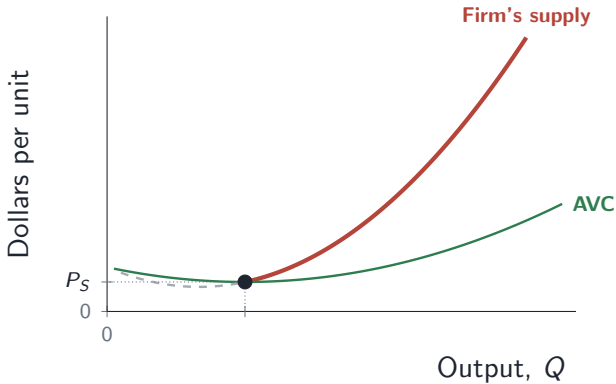
The Short-Run Supply Curve

The firm's supply curve, derived

At each price the firm produces where $P = \text{MC}$.

But it produces only if P is at least minimum AVC.

So its supply curve is the rising part of MC above minimum AVC (P_S). Below P_S it supplies zero.



Example: the supply schedule

Price	Quantity supplied
\$20	0: below minimum AVC (\$24)
\$30	6: 6th MC \$19; 7th MC \$31
\$33	7: 7th MC \$31; 8th MC \$49
\$50	8: 8th MC \$49; 9th MC \$73

Q	VC	TC	MC
0	\$0	\$56	—
1	\$49	\$105	\$49
2	\$80	\$136	\$31
3	\$99	\$155	\$19
4	\$112	\$168	\$13
5	\$125	\$181	\$13
6	\$144	\$200	\$19
7	\$175	\$231	\$31
8	\$224	\$280	\$49
9	\$297	\$353	\$73

Finding

A higher price moves the firm up its MC curve: it supplies more.

From one firm to market supply

Definition: Market supply

The horizontal sum of the firms' supply curves: at each price, add the quantities all firms supply.

100 identical firms, each with the schedule above.

Price	One firm	Market = 100× one firm
\$30	6	$100 \times 6 = 600$
\$33	7	$100 \times 7 = 700$
\$50	8	$100 \times 8 = 800$

In the short run the number of firms is fixed, so market supply slopes up because each firm's MC slopes up.

Part 7

Entry and Exit

Entry and exit change market supply

Assume identical firms, free entry and exit, and unchanged input prices.

Positive economic profit

- New firms enter.
- Market supply shifts right.
- The price falls and profit shrinks.

Persistent economic losses

- Some firms exit.
- Market supply shifts left.
- The price rises and losses shrink.

Finding

Entry and exit stop at zero economic profit: $P =$ minimum ATC.

Zero economic profit

Definition: Zero economic profit

Total revenue covers all economic costs, including the opportunity costs of the owner's time, capital and other resources.

- The salary given up to run the business is part of economic cost.
- Accounting profit can be positive when economic profit is zero.
- The owner earns the normal return available elsewhere. There is no extra economic profit to attract entry.

Part 8

Firm and Industry Equilibrium, and Changes to Equilibrium

Short-run and long-run equilibrium

Definition: Short-run equilibrium

Market quantity demanded equals quantity supplied. Each firm maximizes profit at the market price. Firms can make profits or losses.

Definition: Long-run equilibrium

With identical firms, free entry and exit, and unchanged input prices: firms earn zero economic profit and produce at efficient scale.

Definition: Efficient scale

The output at which average total cost is lowest.

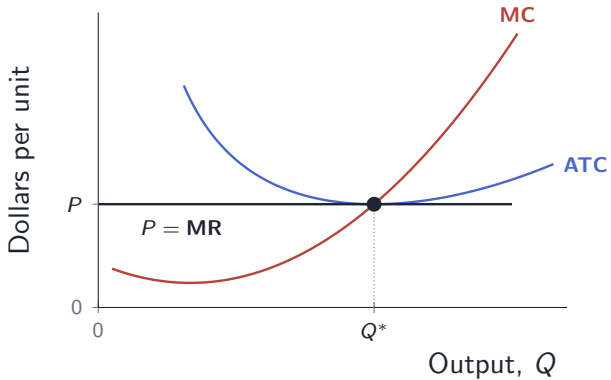
The firm in long-run equilibrium

$P = \text{MR}$: price taking.

$\text{MR} = \text{MC}$: the output rule.

$P = \text{ATC}$: zero economic profit.

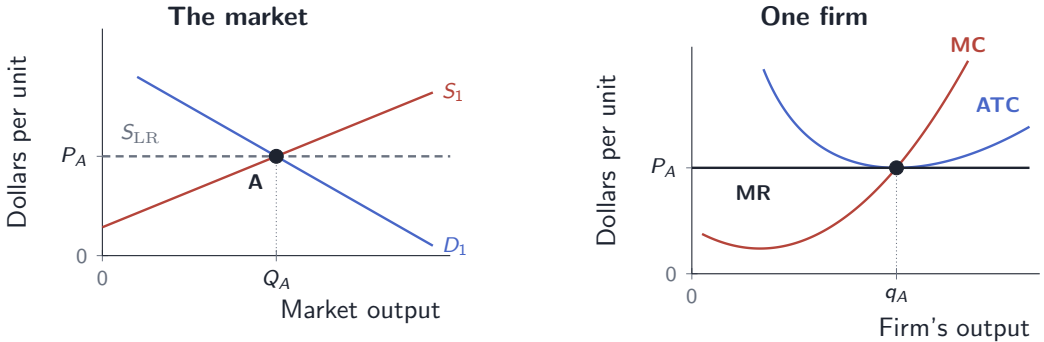
MC meets ATC at its minimum: the efficient scale.



Finding

$P = \text{MR} = \text{MC} =$ minimum ATC. Revenue just covers total cost: $\text{TR} = P \times Q^* = \text{ATC} \times Q^* = \text{TC}$.

Market and firm at A

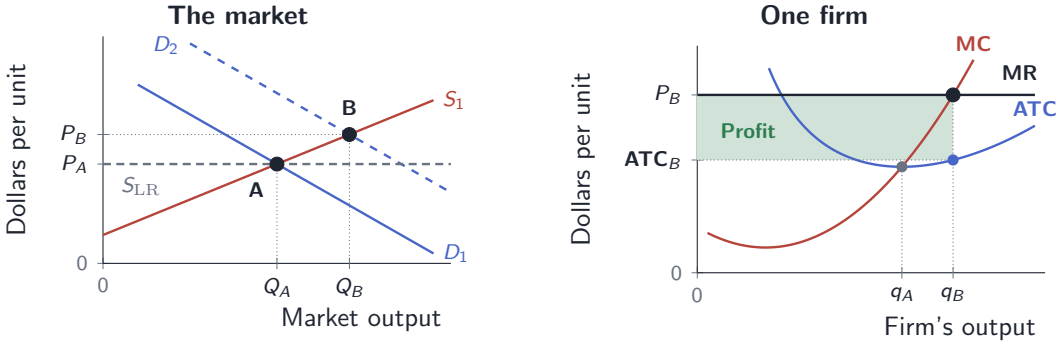


Finding

At A, price equals minimum ATC. Each firm produces at efficient scale and earns zero economic profit. S_{LR} is long-run market supply.

These are schematic panels: compare the direction of changes, not their size.

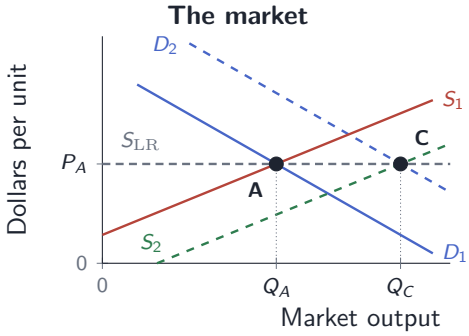
Market and firm at B

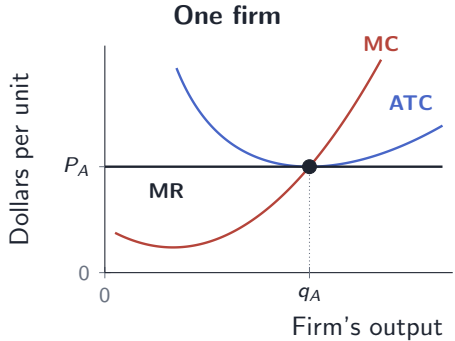


Finding

Demand rises. With the number of firms fixed, price and each firm's output rise.
Profit rectangle: height $P_B - \text{ATC}_B$; width q_B; area $(P_B - \text{ATC}_B) \times q_B$.

Market and firm at C





Finding

Profit attracts entry: supply shifts from S_1 to S_2. Price returns to P_A.
Each firm returns to q_A and zero economic profit. Market output rises from Q_A to Q_C because there are more firms.

A fall in market demand

- In the short run, demand shifts left and the market price falls.
- Each operating firm reduces output along its MC curve.
- All, some, or no firms shut down: this depends on whether the new price falls below their minimum AVC.
- If losses persist, some firms exit. Market supply shifts left and price rises.

Finding

With identical firms and unchanged input prices, exit returns price to minimum ATC. There are fewer firms, each at its original efficient scale.

When long-run supply slopes up

- A scarce input: industry expansion bids up its price. Firms' costs rise. More output requires a higher market price.
- Different costs: the lowest-cost firms enter first. More output requires higher-cost firms, which enter only at a higher price.

Finding

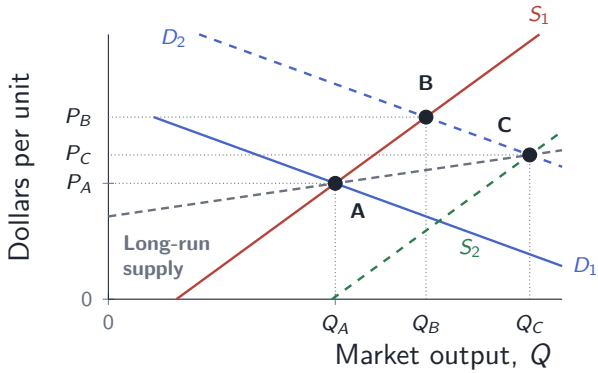
Free entry does not require horizontal long-run market supply. Long-run supply can slope upward.

An upward-sloping long-run supply curve

A: the initial long-run equilibrium.

B: demand rises. With the number of firms fixed, price rises along S_1.

C: entry shifts supply to S_2. Costs have risen, so price remains above P_A.



Finding

Entry lowers price from P_B to P_C. Higher costs keep P_C above P_A.

The marginal firm

Definition: Marginal firm

The firm that would exit in the long run if the price fell any lower. At the current price it earns zero economic profit.

- When costs differ, the market price equals the marginal firm's minimum ATC.
- Lower-cost firms can retain an economic profit if entrants cannot copy their cost advantage.
- Entry stops when no remaining entrant can earn a positive economic profit.

Why competition is efficient

- The market price measures what the last unit is worth to buyers.
- Each firm chooses $P = \text{MC}$: the last unit's value equals its cost.
- Producing more would cost more than buyers are willing to pay. Producing less would give up worthwhile trades.

Finding

With no external costs or benefits, the competitive market produces the efficient quantity.

Review

Practice and Summary

Practice 1: the firm's demand curve

Which change moves a competitive firm's horizontal demand curve upward?

(A) The market price rises.

(B) The firm's fixed cost rises.

(C) The firm increases its own output.

(D) The firm's average total cost rises.

Practice 1: answer

Answer: (A) The market price rises.

Finding

The firm's demand curve is horizontal at the market price. A higher market price moves that line upward.

- A cost change affects profit or supply, not the price buyers will pay this firm.
- This firm's output is too small to move the market price.

Practice 2: a table of total costs

A competitive firm sells at \$9 per unit. How many units maximize profit?

(A) 2

(B) 3

(C) 4

(D) 0: it should shut down.

Q	TC
0	\$6
1	\$10
2	\$16
3	\$24
4	\$34

Practice 2: answer

Answer: (B) 3

1. MC of each unit: $10 - 6 = \$4$; $16 - 10 = \$6$; $24 - 16 = \$8$; $34 - 24 = \$10$.
2. The 3rd unit costs \$8, below \$9: make it. The 4th costs \$10, above \$9: do not.
3. Profit at 3: $27 - 24 = \$3$. At 2: $18 - 16 = \$2$. At 4: $36 - 34 = \$2$.

Q	TC
0	\$6
1	\$10
2	\$16
3	\$24
4	\$34

Practice 3: profit from average total cost

A competitive firm sells 50 units at \$20 each. At 50 units its average total cost is \$16. Its economic profit is

(A) \$36.

(B) \$200.

(C) \$800.

(D) \$1,000.

Practice 3: answer

Answer: (B) $200

Finding

$\pi = (P - \text{ATC}) \times Q = (20 - 16) \times 50 = \$200$.

\$800 is total cost (16×50). \$1,000 is total revenue (20×50). \$36 adds price and ATC, which measures nothing.

Practice 4: keep operating?

A competitive firm's revenue this month would be \$90,000. Its variable costs would be \$100,000, and its fixed costs of \$40,000 must be paid whatever it does. In the short run the firm should keep operating.

True or false?

Practice 4: answer

Answer: False.

- Operate: $90{,}000 - 100{,}000 - 40{,}000 = -\$50{,}000$.
- Shut down: $-\$40{,}000$.
- Revenue does not cover variable cost (TR $<$ VC), so shutting down loses \$10,000 less.

Practice 5: a lasting increase in demand

A competitive market starts in long-run equilibrium. Firms have identical costs, entry is free, and input prices do not change as the industry grows.

> **Finding**
> After a lasting increase in demand, long-run market output is higher even though each firm returns to its original output.

True or false?

Practice 5: answer

Answer: True.

1. Higher demand initially raises price, each firm's output and profit.
2. Profit attracts entry. More firms shift market supply to the right.
3. Price returns to minimum ATC. Each firm returns to efficient scale.
4. There are more firms, so market output is higher.

Which rule answers which question

Question	Rule
What price does one firm face?	The market price: $P = \text{AR} = \text{MR}$.
How much should it produce?	$\text{MR} = \text{MC}$ on rising MC, if it operates.
Does it make a profit?	$\pi = (P - \text{ATC}) \times Q$.
Should it shut down now?	Shut if $P < \text{AVC}$; threshold: minimum AVC.
Should it stay in the long run?	Exit if revenue cannot cover all avoidable costs.
What changes market supply?	Entry shifts it right; exit shifts it left.
Where does entry and exit stop?	$P =$ minimum ATC with identical firms and unchanged input prices.